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Vector System Manual

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Last Revised on August 14, 2020 1:24 PM

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Introduction

This service manual has been developed to provide technicians with the basic understanding of the Vector Fuel System manufactured by Optimus Technologies. This manual should be used in conjunction with the base engine manual and the OEM service manual. Any maintenance or repairs should be performed by trained and experienced service technicians.

Any installations performed that are not in accordance with this manual and service procedures, or those performed by technicians not certified by Optimus will void the warranty. No attempt should be made to operate the system until this manual and supporting documents have been read and fully understood.

Disclaimer

All information, illustrations and specifications contained in this manual are based upon the latest product information available at the time of publication. Optimus will make its best efforts to update clients of manual changes. New products or upgrades introduced after the issue date of this manual will be added as supplements or updated revisions. The latest information is available through Optimus or its distributors.

Contact Information

For the latest manual or any questions regarding content, please contact Optimus Technologies at 412.727.8228 or email info@optimustec.com.

General Safety

When performing maintenance or installation work on a vehicle, use standard safety practices and common sense. Wear safety glasses when operating power tools such as drills, disconnect the battery cables before routing electrical lines, and follow all safety protocols listed in the OEM manual. With many variations in procedures, tools, and service parts, Optimus cannot state all possible safety conditions and hazards.

Disregard for warnings, cautions, and instructions can lead to injury, death, or damage to the engine or vehicle.

Due to the close proximity of components on the vehicles, follow all local guidelines with regards to COVID-19 required PPE and safety precautions when installing the Vector System.

⚠ Before working on a vehicle, set the parking brakes, place the transmission in neutral, and chock the wheels. Failure to do so can result in unexpected vehicle movement and can cause serious personal injury or death.

⚠ Exhaust gases contain carbon monoxide. Always run the engine outdoors or use a properly vented exhaust hose. Prolonged or excessive exposure may cause serious illness or death.

⚠ Never operate the engine in an area where hydrocarbon vapors (gasoline, for example) are present or are suspected to be present. Hydrocarbon vapors can enter the air intake and over speed the engine, causing severe engine damage and/or an explosion and fire. Serious personal injury or death could occur.

⚠ Do not attempt to repair or service this vehicle without having sufficient training, correct service literature and the proper tools. Failure to follow this could lead to personal injury or making your vehicle unsafe.

⚠ Diesel engine exhaust and some of its constituents are known to the state of California to cause cancer, birth defects and other reproductive harm.

⚠ Wear appropriate apparel and protective eye wear when working around hot engines and components.

⚠ Make sure all tools and power cords are in good condition and ensure that all standard electrical tools are grounded.

Tools

A full set of diesel-mechanics tools is recommended to perform an installation. Some specific tools you may need include:

- Cummins fuel line disconnect tools.
- Duetsch barrel crimp tool.
- Deutsch terminal removal tool; plastic
- Remote hose clamp pliers

Fuel Specifications

All fuels used should comply with California Air Resources Board (CARB) regulations. CARB Diesel requires D975 standard specification minimums with additional limits on sulfur, aromatics, and lubricity. The following table displays minimum diesel fuel specifications.

Property	CARB Diesel Value
Flash Point, °C, min.	52
Water and Sediment, % vol	0.05
Distillation Temperature, °C90, % vol	-
min	282
max	338
Kinematic Viscosity, mm^2/S at 40°C	-
min	1.9
max	4.1
Ash % mass, max	0.01
Sulfur, ppm (µg/g) max	15
Copper strip corrosion rating, max	No. 3
Cetane number, min	40*
Aromaticity, % vol, max	10
Ramsbottom carbon residue on 10 % distillation residue, % mass, max	0.35
Lubricity, HFRR @ 60°C, micron, max	520
Conductivity, pS/m, min	25

* A minimum cetane rating of 40 is acceptable provided emissions are equivalent to fuel rated at 48 cetane and 10% aromaticity volume.

Biodiesel

For biodiesel blends, the Optimus System requires a minimum fuel quality of ASTM D6751 Grade No. 1-B specifications with the following substitutions.

Property	B100 No. 1-B Value
Water and Sediment, % vol max	0.05
Oxidation Stability (110 °C), hours	6
Sodium & Potassium combined, ppm	-
Calcium & Magnesium combined, ppm max	-
Sodium, Potassium, Calcium, & Magnesium combined, ppm max	4*
Phosphorus, ppm max	4*

* 4 ppm is acceptable per individual fuel load; however, the weighted average for all biodiesel utilized must be 2 ppm or less.

Vector System General Overview

The Vector System is an advanced fuel supply system that enables the engine to operate from two independent fuel supplies. For purposes of this installation manual, the OEM fuel supply system is referred to as the Primary System, and the Optimus Vector System is referred to as the Vector System.

Operation Overview

At vehicle startup, the Vector System begins a series of cycles and processes that ensure reliable, safe, and efficient performance when using different fuel types. A combination of hardware and software is initiated by the vehicle's keyed ignition without any additional input from the operator.

The vehicle starts using the Primary System. Meanwhile, the Vector System enters a warm-up cycle in which waste engine heat via the coolant system begins circulating through the Vector Manifold and Vector Fuel Tank. When predefined temperature and pressure conditions are met, the Vector System engages, and the engine is supplied solely by the fuel in the Vector Tank.

Shutting down the vehicle initiates a purge cycle that re-engages the Primary System. This ensures that the vehicle is ready for its next operation cycle. When that process finishes, the vehicle automatically powers down.

1. **System State 1 – OEM Mode**
   - Vehicle fueled by Primary System
2. **System State 2 – Warm-Up Mode**
   - Vehicle Prepares for Switch to Vector System
3. **System State 3 – Vector Mode**
   - Vehicle Fueled by Vector System
4. **System State 4 – Purge Mode**
   - Key Removed, Fuel Lines Purged

Operating State One - OEM Mode

Upon vehicle startup, a number of automatic processes are initiated to ensure fuel in the Vector Tank is appropriate for use. The Optimus Electronic Control Unit (ECU) activates a user interface located in the cab of the vehicle that displays operating status, Vector Fuel Tank level, and any warnings or maintenance requirements.

State 1 - OEM Mode	
Fuel Pump:	Off
Circulation Valve:	Off
Fuel Supply Selector Valve:	Off
Fuel Return Selector Valve: (If Applicable)	Off

Operating State Two - Warm-Up Mode

Warming fuel is directed first from the Vector Tank to the Vector Manifold, and then to a fuel supply selector valve in the engine. At this point, the supply valve is in a de-energized/closed position as the vehicle is still powered by the Primary System. Fuel continues to warm as it is redirected back to the Vector Manifold through a recirculation valve.

State 2 - Warm-Up Mode	
Fuel Pump:	On
Circulation Valve:	On
Fuel Supply Selector Valve:	Off
Fuel Return Selector Valve:	Off
(If Applicable)	

Operating State Three - Vector Mode

With all operating conditions met, the Vector System begins to supply fuel from the Vector Tank to the engine. During this transition, the Optimus ECU deactivates the circulation valve in the Manifold, which builds pressure in the fuel system. The fuel supply selector valve actuates and allows fuel from the Vector Tank to flow to the engine. Concurrently, the flow of fuel from the Primary System is ceased or redirected.

State 3 - Vector Mode	
Fuel Pump:	On
Circulation Valve:	Off
Fuel Supply Selector Valve:	On
Fuel Return Selector Valve:	On
(If Applicable)	

If applicable, a fuel return selector valve will actuate on a time delay after the fuel supply valve activates. This delay allows for the evacuation of fuel from the shared lines.

The Vector Mode lasts the vast majority of vehicle runtime. If any operating conditions are not met, the Optimus ECU automatically returns to OEM mode.

Operating State Four - Purge Mode

Before vehicle shutdown, the Vector System deactivates the fuel selector valves that resumes operation of the Primary System. The Optimus ECU automatically begins this purge cycle once the vehicle operator removes the key, and the vehicle will continue to run until its completion approximately 90-180 seconds later.

State 4 - Purge Mode	
Fuel Pump:	Off
Circulation Valve:	Off
Fuel Supply Selector Valve:	Off
Fuel Return Selector Valve:	Off
(If Applicable)	

In the event of an emergency, the vehicle operator can bypass the purge cycle by pushing the Safety Stop Button on the user interface. This will immediately end the purge cycle, shut off all electrical and mechanical components, and shut down the vehicle.

Vector System Components

Vector Fuel Tank

The Vector Fuel Tank is an additional fuel tank or tank compartment designed to hold fuel to be processed by the Vector System. The tank contains a cross flow heat exchanger that transfers heat from the engine coolant to the stored fuel.

The tank must meet all applicable Federal Motor Carrier Safety Administration regulations and comply with all OEM requirements. An aluminum tank is preferred, but other materials suitable for fuel storage can be utilized.

Depending on vehicle specifications, there are a variety of tank mounting options, such as two separate tanks in a saddle configuration, or one split tank with two compartments.

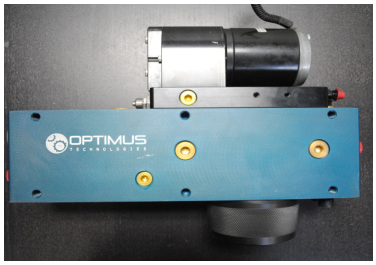
Vector Fuel Tank
(Split Tank Configuration)

In-Tank Heat Exchanger

Vector Manifold

The Vector Manifold processes fuel from the Vector Tank to ensure it is suitable for injection. The manifold contains the following components:

- a liquid-liquid cross flow heat exchanger that utilizes heated coolant from the engine to further elevate fuel temperature
- electric fuel pump
- fuel pressure regulator
- 150-micron cleanable pre-filter (vacuum side)
- disposable primary filter cartridge (pressure side)
- fuel-air separation system, with fuel-air return-to-tank port and a cleanable fuel-settling bowl with water drain.



Vector Manifold

The fuel system pressure and flow specifications are designed to meet OEM fuel delivery specifications. The Vector System utilizes fuel filters that are compatible with the Primary System for each engine application.

There are sensors monitoring fuel inlet and outlet temperatures, the fuel pressure pre-filter, the fuel pressure post-filter, and the coolant inlet and outlet temperatures. The Optimus ECU displays these all of these parameters and readings.

Fuel Selector Valves

The Vector System utilizes directional control valves to select fuel flow pathways depending on a number of operational conditions. There are a variety of possible fuel valve configurations which all perform the same functions in a slightly different manner. The valve system utilized is engine specific and is designed to maintain the functionality of the OEM fuel and emission control systems when operating on either OEM or Vector fuel.

Fuel Selector Valves

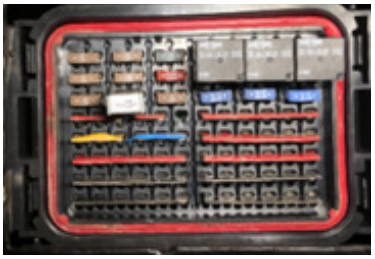
Electronics Control Unit (ECU)

The Optimus ECU is an electrical hardware component that is mounted to the vehicle. The ECU automates all aspects of the system by using collected information to determine the appropriate operating state.

ECU Installed on Vehicle

Power Distribution Module (PDM)

The PDM is a rugged, waterproof enclosure which houses the fuses, relays, diodes, and busbars which distribute power throughout the Vector System.



Power Distribution Module

Telematics Unit

The telematics platform enables remote tracking and management of assets through GPS and cellular data technology. This system serves as a gateway for remote administration of vehicles and data analysis. See page 16 for installation instructions.

C4MAX Telematics Unit

User Interface

The primary function of the user interface is to provide the operator with status updates for the Vector System. The interface displays the Vector Tank level and the system's current operational status. The interface also conveys any system error or warning messages.

Three context sensitive buttons navigate through the menus of the interface. After startup, hold the middle button to see system warnings or messages.

User Interface in Dash

User Interface Icons

Icon	Description	Additional Details
	Low Vector Tank Fuel Level	System will revert to primary fuel once depleted
	Change Fuel Filter	Regularly scheduled fuel filter maintenance due
	Low Fuel Pressure	Contact Optimus or Service Provider
	Diesel Particulate Filter Regeneration in Progress	
STOP	Emergency Stop Icon	Stops All Electrical and Mechanical Components